



Properties and Applications of Non-Ferrous Metals 2

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Non-Ferrous Metals

Comparison of properties:

Alloy system	Examples	Elastic mod GPa	Density kg/m ³	Melting T °C	Notes
Ferrous	Steel	200	7850	1536	Corrosion
Light metals	Al	70	2700	660	Low density Low T _m
	Mg	45	1700	650	
	Ti	110	4500	1660	High specific strength
Transition metals	Cu	120	8900	1083	Good corrosion resistance
	Ni	210	8900	1453	
Refractory metals	W	400	19300	3410	Very high T _m
	Mo	329	10200	2610	
Noble metals	Au	80	19300	1062	No corrosion
	Ag	76	10500	962	
	Pt	170	21400	1770	

Copper alloys

Unalloyed Pure Copper – high conductivity

- Electrical and thermal applications

Types of copper alloys:

- **Brass** – alloy with zinc (Cu-Zn), good corrosion resistance
 - Bolts, pipe fittings, taps
- **Bronze** – alloy of copper and several other elements including Sn, Al, Si, and Ni (but usually known Cu-Sn), corrosion and fatigue resistance
 - Marine applications, bearings, bushings
- **Beryllium Copper** (Cu-Be), very high yield strength
 - Springs, jet aircraft landing, gear bearings and bushings₃

Copper alloys

Pure Copper – high purity, oxygen free

Low electrical resistivity = 1.7×10^{-8} Ohm.m

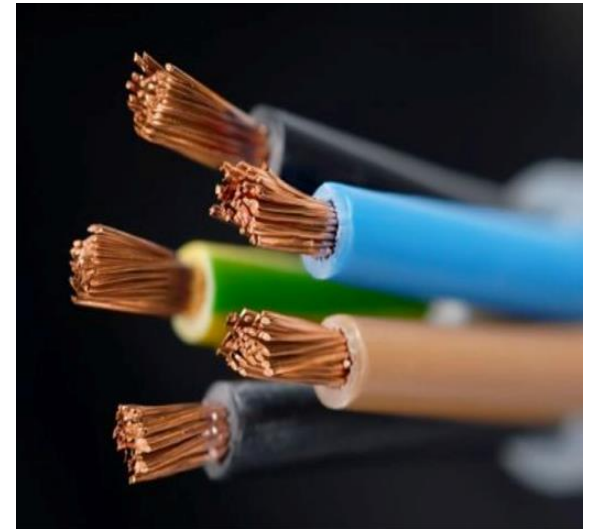
High thermal conductivity = 400 W / mK

Ductile metal, easily deformed and shaped

Applications:

Electrical wiring

Heat exchangers

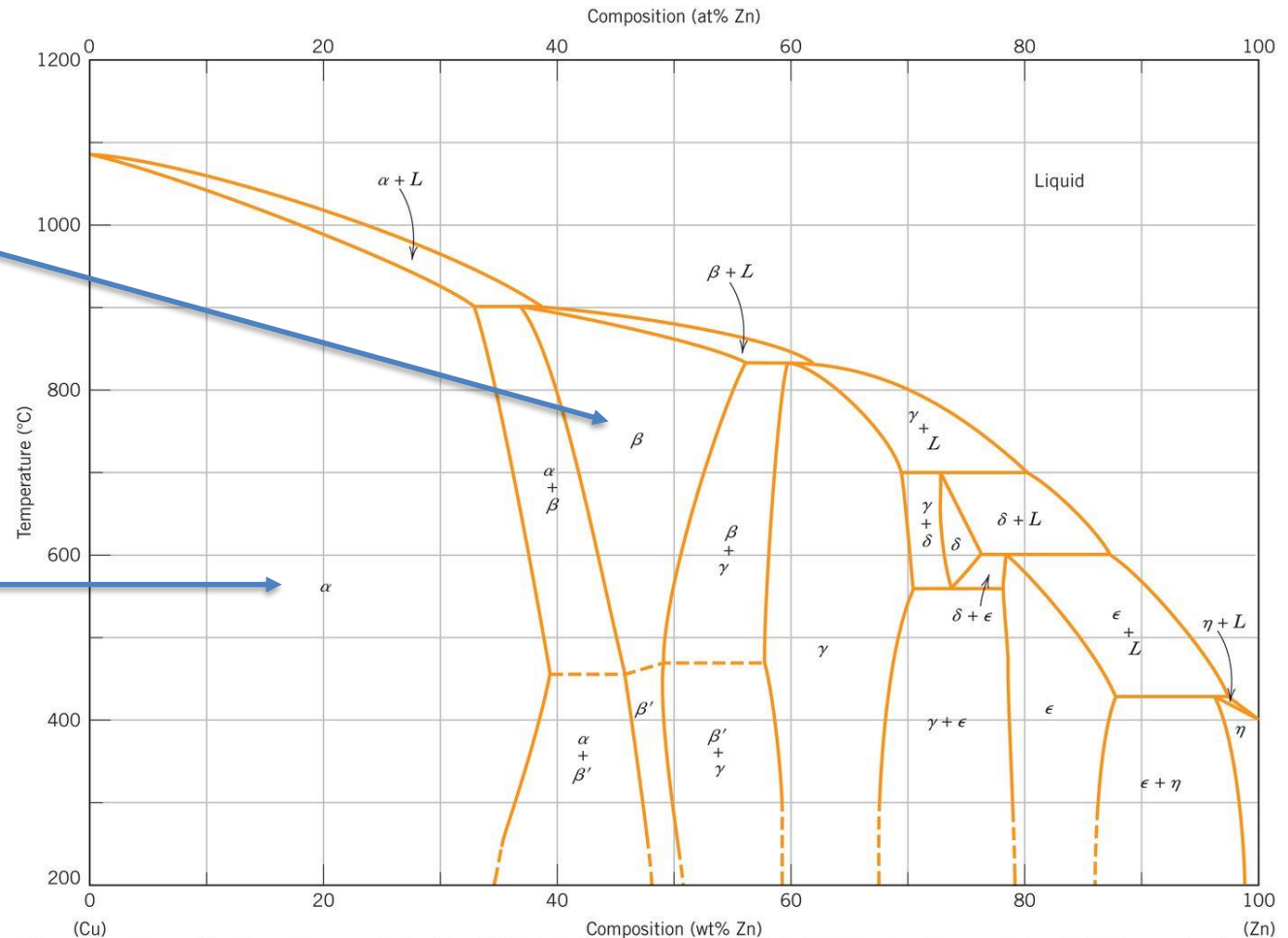


Copper alloys

Brass – alloy of copper and zinc

bcc β phase

fcc α phase



Adapted from Binary Alloy Phase Diagrams, 2nd edition, Vol. 2, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.

Copper alloys

Brass – alloy of copper and zinc (Zn)

α phase has the fcc crystal structure
ductile and easily cold worked

β' phase has the bcc crystal structure
harder and stronger phase

Alloys of $\alpha + \beta'$ have to be hot worked

Some common brasses are yellow, naval, and cartridge brass

Applications:

Costume jewelry, musical instruments, coins, electronic packaging

Corrosion resistant bolts, Water pipe fittings



Copper alloys

Bronze – alloy of copper and tin (Sn) + other alloying elements

Cu – Sn alloys are stronger than brass alloys

They have good corrosion resistance (especially to sea water)

They have good fatigue resistance

They have good tensile properties

Applications:

Bearings

Bellows

Propellers on ships



Copper alloys

Beryllium copper – alloy of copper and beryllium (Be)

Copper alloys with 0.4 – 2.0 wt% Be

Strengthened by heat treatment (precipitation hardening)

Very high tensile strengths = 1400 MPa

Beryllium dust is very harmful (toxic)

so processing of these alloys has to be done in controlled conditions

Applications:

Springs and clips

Bellows

Electrical contacts and switches

Copper alloys

Compositions and Mechanical Properties of copper alloys:

Alloy	Composition (wt%)	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
Electrolytic	Pure Cu	Annealed	220	69	45
Catridge Brass	Cu 30Zn (α)	Annealed	300	75	68
		Cold worked	525	435	8
Yellow brass	Cu 29Zn 3Pb	As cast	234	83	35
Phosphor bronze	Cu 5Sn 0.2P	Annealed	325	130	64
		Cold worked	560	515	10
Tin bronze	Cu 10Sn 2Zn	As cast	310	152	25
Beryllium copper	Cu 1.9Be	Precipitation hardened	1140 - 1310	690 - 860	4 - 10

Nickel alloys

Nickel alloys – **heat treatable**

Nickel metal – corrosion resistant, especially to alkalis

fcc crystal structure

High melting temperature = 1453°C

Important alloys:

Monel = Ni 28%Cu 7%Fe

Inconel 625 = Ni 22%Cr 9%Mo 3.5%Nb

Applications:

High temperature corrosive environments

Exhaust gas systems, chemical processing

Pumps

Nickel super alloys

Properties of nickel super alloys – **heat treatable**

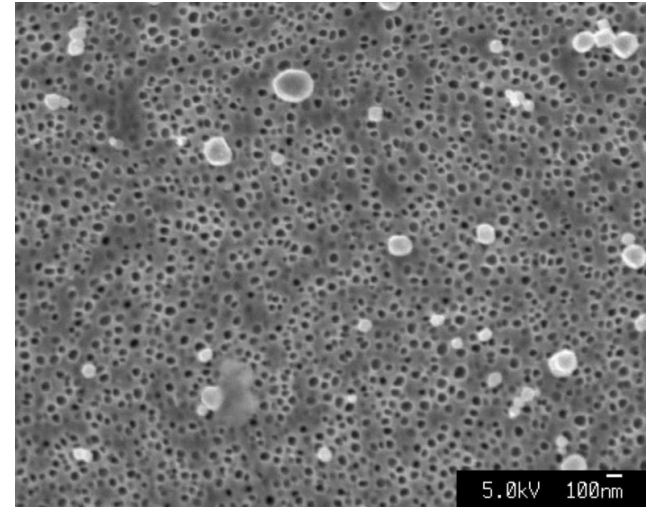
Alloys of Ni with Cr, Ti and Al
+ other elements Co, Mo, W
e.g. Ni-9.7Al-1.7Ti-17.1Cr-6.3Co-2.3W at%

Their extraordinary properties come from
the γ' phase Ni_3Al or Ni_3Ti as very small
(50nm) particles in the Ni matrix

These alloys are resistant to creep failure
Deformation at high temperature $> 850^\circ\text{C}$

Applications:

Aircraft – turbine blades



Refractory metals

The refractory metals have very high melting temperatures

Tungsten W = 3410°C

Tantalum Ta = 3017°C

Molybdenum Mo = 2623°C

Niobium Nb = 2468°C

Their high T_m makes these materials very difficult to process

They also have high density

Applications:

Material processing, dies and tools

High temperature applications

Noble metals

The noble metals have the property of not forming an oxide

Gold Au – good electrical and thermal conductivity = 300 W/mK

Silver Ag – 100% reflectivity

Platinum Pt – high melting temperature = 1768°C

Applications:

Chemical processing

Space applications – mirrors, radiation shields

Gold – electrical contacts in micro-electronics

Medical applications – X-ray markers, dental materials

Summary

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